TOPIC 4.1

What is the universe, and how do we make sense of it?

Key Concepts

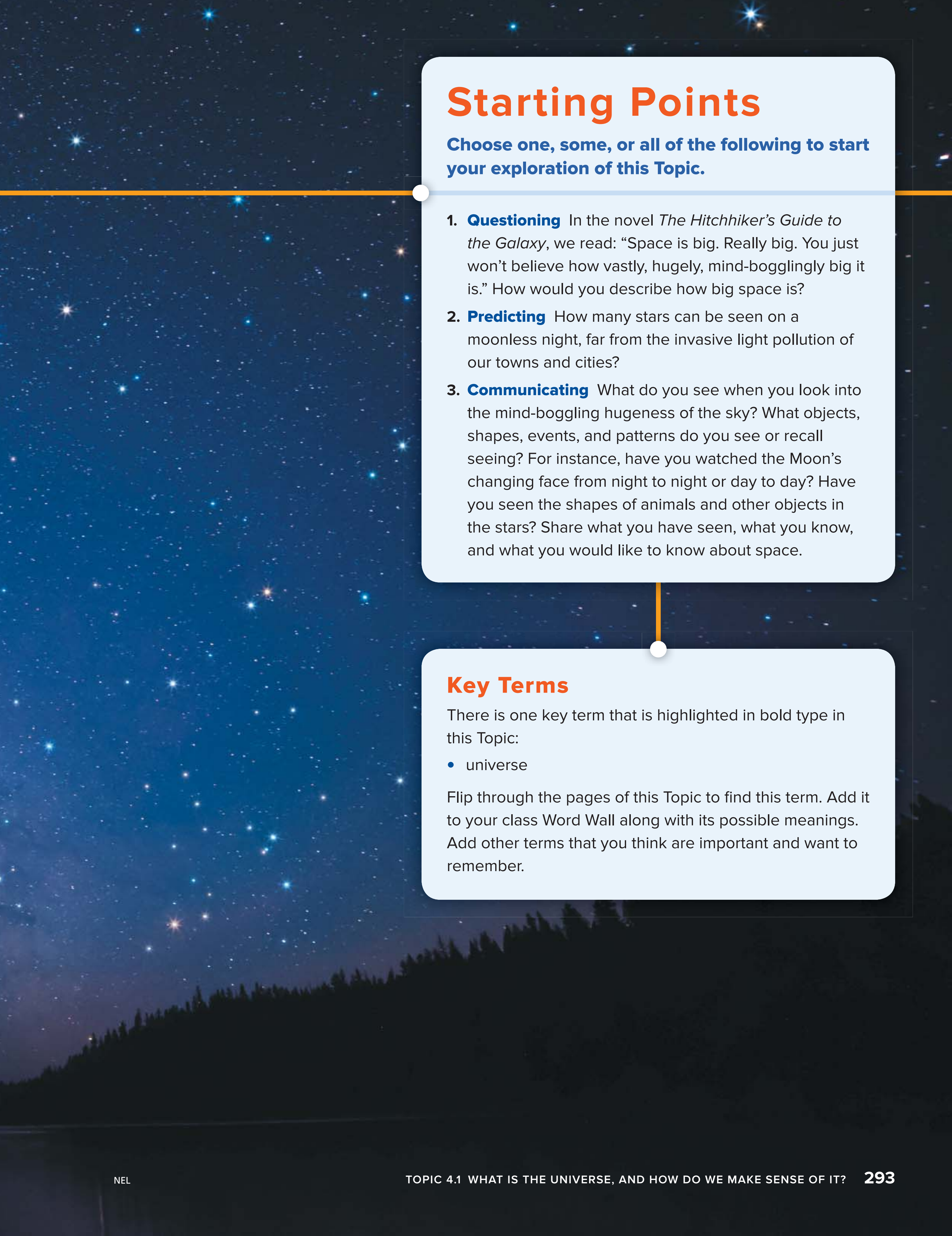
- "Universe" is understood as different things to different people.
- The sense we make of the universe depends on who we are, when we are, and what we know.

Curricular Competencies

- Make observations aimed at identifying questions about the natural world
- Experience and interpret the local environment
- Apply First Peoples perspectives and knowledge, other ways of knowing, and local knowledge as information sources
- Demonstrate an awareness of assumptions and bias
- Express and reflect on experiences, perspectives, and worldviews through place

You have many ways to describe and share what you see in the sky. Imagine sharing your ideas about the sky with friends, and they pass this information to others. Imagine this sharing takes place over months, years, centuries, and longer. This process of sharing is the act of storytelling. For millennia, the sky has inspired us to weave stories, to pass on ideas, explanations, and wisdom. We tell:

- ancient stories about shapes, patterns, and events our ancestors saw
- scientific stories to link observations with hypotheses and theories
- etched and painted stories on rock, hides, wood, paper, and pixels, to communicate our sense of wonder and connection
- visionary stories—written, spoken, sung, and danced—to amuse and teach
- inspirational stories from Elders, seers, and spiritual leaders to link space, Earth, and all creation in our hearts, minds, and spirits



Starting Points

Choose one, some, or all of the following to start your exploration of this Topic.

1. **Questioning** In the novel *The Hitchhiker's Guide to the Galaxy*, we read: "Space is big. Really big. You just won't believe how vastly, hugely, mind-bogglingly big it is." How would you describe how big space is?
2. **Predicting** How many stars can be seen on a moonless night, far from the invasive light pollution of our towns and cities?
3. **Communicating** What do you see when you look into the mind-boggling hugeness of the sky? What objects, shapes, events, and patterns do you see or recall seeing? For instance, have you watched the Moon's changing face from night to night or day to day? Have you seen the shapes of animals and other objects in the stars? Share what you have seen, what you know, and what you would like to know about space.

Key Terms

There is one key term that is highlighted in bold type in this Topic:

- universe

Flip through the pages of this Topic to find this term. Add it to your class Word Wall along with its possible meanings. Add other terms that you think are important and want to remember.

CONCEPT 1

“Universe” is understood as different things to different people.

Some stories begin in the present and move forward into the future, or they begin in the present and they flash back to the past. Some stories begin *in medias res*, which is a Latin term meaning “in the middle.” Such stories are often epic tales like the *Iliad* of Homer, the first Star Wars film (which is Episode IV in the complete saga), and the video game *Final Fantasy X*.

Many stories, however, begin at the beginning. It might be the beginning of an action or event, or the beginning of the life of a character we will come to know more about, or some mixture of these.

If a scientist were to begin a story about the universe at the beginning, this story would take us back to a time approximately 13.8 billion years ago. See **Figure 4.1**.

Take a moment to think about that.

Many people find it a challenge to imagine the size of a number like 10 000. Even more find it a challenge to imagine a number as large as 100 000. Few people can fathom what 1 million looks like, let alone 1 billion. So what (on Earth? beyond Earth?) is the size of 13.8 billion?

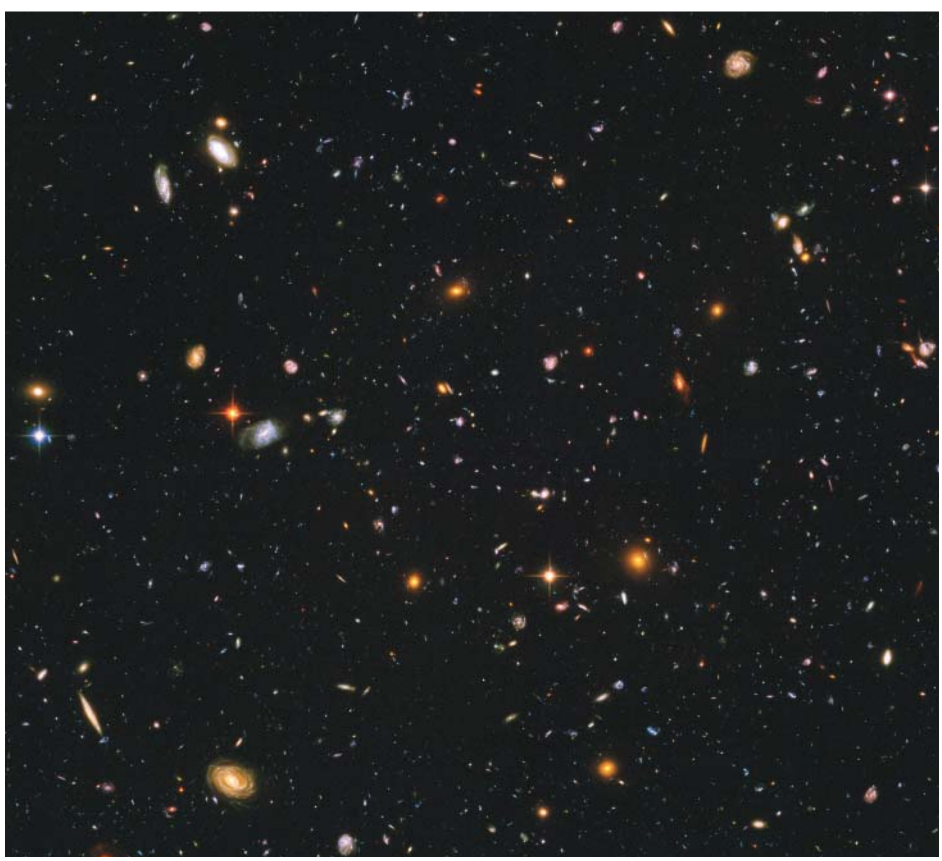


Figure 4.1 This photo captures galaxies (collections of stars) at many different stages of their lives. Some are very young—barely 1 billion years old. It has taken their light almost 13 billion years to reach the telescope that photographed them. The area of sky represented by the photo, called the Hubble Ultra Deep Field, is tiny. Imagine a dime 23 m away from you. That dime is the size of the area of sky visible to the telescope.

Activity

Bringing Big Numbers Down to Earth

Consider the size of the numbers in each of the following. What does that number look like? How can you make it easy or easier to grasp? For example, could you develop an analogy? Could you make a scale model? Could you write computer code that generates that number of X's on a screen? What other ideas do you have to bring large numbers "down to Earth?"

- Gabriel has 250 followers on social media.
- Jen's comic book collection has reached nearly 2500.
- The tallest mountain peak in British Columbia is Mount Waddington, in the southwest of the province, at a little more than 4000 m.
- The deepest part of Earth's ocean is the Mariana Trench, east of the Philippines, at nearly 11 000 m.
- Data from the census conducted in 2016 showed that the city of Victoria had a population of nearly 86 000 people.
- Data from the 2016 census showed that BC's population was nearly 4.7 million.
- At the time this book was published, Earth's human population was more than 7.6 billion, and counting.
- A recently discovered lobster species, named *Yawunik kootenayi* in honour of the Ktunaxa of southeastern BC, lived about 500 million years ago.
- Add one more number of your own, as large as you want. Go ahead—reach for the stars!

Fossil of *Yawunik kootenayi* from Marble Canyon in Kootenay National Park. Yawunik is a key figure in the Ktunaxa creation story. The fossil now resides at the Royal Ontario Museum.

What Does It Mean to Talk about the Beginning of the Universe?

Talking about the beginning of the universe raises a host of questions. For example:

- What happened that made the universe begin?
- Was there anything that existed before the universe began?
- What happened after the universe began?
- How do we know the universe began about 13.8 billion years ago? Could it have been earlier? Could it have been later?
- Is there only one universe?
- Is it possible that the universe has begun more than once?
- Is it possible that the universe didn't begin at all? What could that even mean?

Take a moment. (Yes, another moment. After 13.8 billion years, we have time.) What questions of your own—about the universe, about time, about big numbers, about this unusual textbook discussion—do you have? As you pose and discuss your questions, consider one more, which might or might not be a simple one: What does the word "universe" mean in the first place? We'll pick up with this question when you turn the page.

Reflecting on the Meaning of "Universe"

universe all that exists everywhere, including all matter, energy, planets, stars, galaxies, and the space in which all of this exists

Consult any dictionary, and you will likely see **universe** defined as "all there is" or "all that exists." It is possible to add details about what "all" could mean, as in the definition in the margin. Does that definition seem reasonably simple and straightforward to you? If so, consider this. We know that there are billions of galaxies (collections of stars). However, barely 100 years ago, most scientists believed that the Milky Way galaxy—the galaxy in which our planet is found—was the *whole* universe. In other words, the universe was all that exists in the Milky Way!

This is like saying that a person living in a small room believes that the universe is everything in that room and the room itself. Imagine what it would be like for that person to step outside that room and witness all the other rooms and buildings and roads and cars and people and animals and clouds and sky—a daytime sky filled with clouds and a great yellow, glowing orb, and a nighttime sky filled with pinpoints of light. Imagine the shock to that person's senses. Imagine trying to shift your understanding to accommodate such a startling, different view of all that you thought before.

The "universe" may well be all that exists, but "all that exists" is relative. This means that your understanding of all that exists depends on who you are, where you live, and the time that you live. In the next Concept, you will begin to think further about this idea.

Activity

Life in the Bowl of the World

The country of Tanzania in Africa has one of the world's great wonders—a deep crater teeming with life, cut off from the surrounding countryside. Imagine that you live in this "bowl" of a world, called Ngorongoro Crater. All you need to sustain your life is found here. Because the rock walls around you are 600 m, your sense of place is well-defined by them.

1. Compare Ngorongoro Crater to the ideas discussed on this page. Imagine living in this place. How would your understanding of your world, the universe, and yourself be affected if you could rise above and see beyond the crater walls for the first time?
2. Is there anything in your own life, or in the life of someone you know or have learned about, that is like the scenario in this activity? Share your thoughts privately or with the class.

Before you leave this page . . .

1. What is your own personal understanding of the word "universe?"
2. In what ways is the meaning of "universe" relative, and how is that significant?

CONCEPT 2

The sense we make of the universe depends on who we are, when we are, and what we know.

Each photo in **Figure 4.2** shows a technology that, at that time, was a major communication device. Would you know how to understand and use the device in Photos A and B if you could step out of Photo C and enter those earlier worlds? What about if the person in Photo A stepped into photo B, or if the people in Photos A and B stepped into Photo C?

The time frame in **Figure 4.2** is only 100 years. What if the gulf of time were much broader? How would you feel if you stepped into the world of 16th century England, the world of 14th century North America, the world of 2nd century China, or any time and place of your choosing? What if someone your age from those periods and places stepped into your classroom right now?

The term culture shock is used to describe the sense of anxiety and disorientation that a person feels when they experience a dramatically new social and cultural situation. What kind of culture shock would you feel? What about your counterparts from those other times? What things would you know that they don't? What things would they know that you don't?

Return to the present, and consider: How does the period of time when you live affect what you think and believe and how you live your life? How does where you live affect these same things? How do all these things affect your understanding of who you are? (And what do you interpret the phrase "who you are" to mean?)

These are big questions, deep questions, essential questions. Ask yourself these questions every so often as you explore this unit.

Figure 4.2 Communications technologies at three moments in time over a period of about 100 years

Before you leave this page . . .

1. Re-read the last paragraph. Do you think your answers to questions like those in the paragraph could change over the course of the unit? Do they apply only to your study of this unit?

TAKE Action

Make a Difference

What should we do about light pollution?

Most people who live in urban centres are unable to see the stars of our own galaxy. Since the invention of artificial lighting, our towns and cities have become polluted by our efforts to light up the night. Night lighting improves safety in cities. It also allows us to work and shop after the Sun sets. However, light pollution disrupts the cycles that govern sleeping, feeding, migration, and breeding of many living things. Birds can get lost flying over cities at night if they cannot see the stars to guide them. Newly hatched sea turtles are normally guided to the sea by starlight and moonlight reflecting off the water. However, bright beach lights can send them in the wrong direction, sometimes into the path of oncoming traffic.

Analyze and Evaluate

1. Who else is affected by light pollution, and in what ways? (Don't forget astronomers.)
2. What solutions have been proposed to reduce or combat light pollution? How effective are they?

Communicate

3. What arguments can be made in favour of having well-lit cities and towns, even at the expense of light pollution?
4. Provide a well-supported argument that expresses your views of light pollution and what, if anything, should be done about it.

Check Your Understanding of Topic 4.1

QP Questioning and Predicting PC Planning and Conducting PA Processing and Analyzing E Evaluating
AI Applying and Innovating C Communicating

Understanding Key Ideas

1. Think about the word universe.
   - a) What is a simple, common-sense definition for this word?
   - b) What observable evidence supports this definition? PA

2. You have probably heard or read the phrase, "It's all relative." This means that how you think about or perceive something depends on its situation or context. In what ways is the definition for universe relative? QP PA

Connecting Ideas

3. The Sun is millions of kilometres from Earth, and yet the Sun and its effects are present in our daily lives in many ways. For example, the food we eat depends on the Sun's energy. The gasoline that runs cars, chainsaws, and snow machines comes from ancient plants and marine algae that originally grew with energy from the Sun.
   - a) Describe two ways that the faraway Sun is present in your daily life.
   - b) The universe (whatever we mean by it) is also far away from us. In what ways is it present in your daily life? Note: Don't limit your thinking to just the physical. How might the universe be present in your life in other ways? Think about First Peoples perspectives in science and other ways of knowing to help you reflect on your responses. E AI

4. Early skywatchers paid close attention to the sky. Why might the sky be important to
   - a) a hunter?
   - b) a farmer?
   - c) a sailor?
   - d) a ruler or chief?
   - e) an Elder or knowledge-keeper? E AI

5. Imagine that you are skywatching on a clear, dark night. You see a view of stars like the one shown in the photo. What questions can you formulate about the following?
   - a) their distances from Earth
   - b) their distances from one another
   - c) their colours and/or brightness
   - d) any other questions that occur to you

Note: The purpose of formulating these questions is to model the brainstorming stages of the process of scientific inquiry. Depending on your level of interest, and in consultation with your teacher, you could choose to further investigate one or more of your questions. QP C

Making New Connections

6. Stories that people tell about the universe and its creation or formation are often stories about Earth (or places on Earth) and its creation or formation. Does that surprise you? Explain why or why not. E AI

7. People who are interested in space and astronomy often have careers or interests in fields other than science. Do you think this is something unique to space and astronomy, or does it apply to any science-related field? Why do you think so? E AI C